



UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	:	DATABASE DESIGN AND DEVELOPMENT
COURSE CODE	:	ICT450/ITS432
EXAMINATION	:	DECEMBER 2019
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of five (5) questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of :
 - i) the Question Paper
 - ii) an Answer Booklet – provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

QUESTION 1

- a) File system method is an improvement over manual system. However, there are many problems that associated with file system including data anomalies.
- i) Briefly describe **TWO (2)** types of data anomalies. (4 marks)
 - ii) Briefly describe **TWO (2)** applications of big data implemented in healthcare industry. Give an example for each application. (4 marks)
- b) Data redundancy is common issue in file system. Illustrate with description how data redundancy could occur. (8 marks)
- c) Based on the Entity Relationship Diagram in Figure 1.0, explain how referential integrity are enforced.

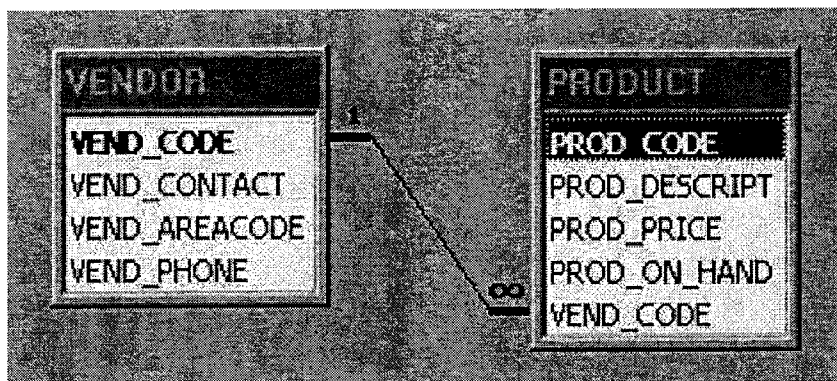


Figure 1.0 : Referential integrity in Entity Relationship Diagram

(4 marks)

QUESTION 2

- a) A proper written business rules are used to define entities, attributes, relationships, and constraints. Careful derivation and definition of business rules are crucial to produce a good database design. Based on Figure 2.0, describe in the context of database design. the concept of entity and attribute. Give **TWO (2)** examples of entities and attributes in your description.

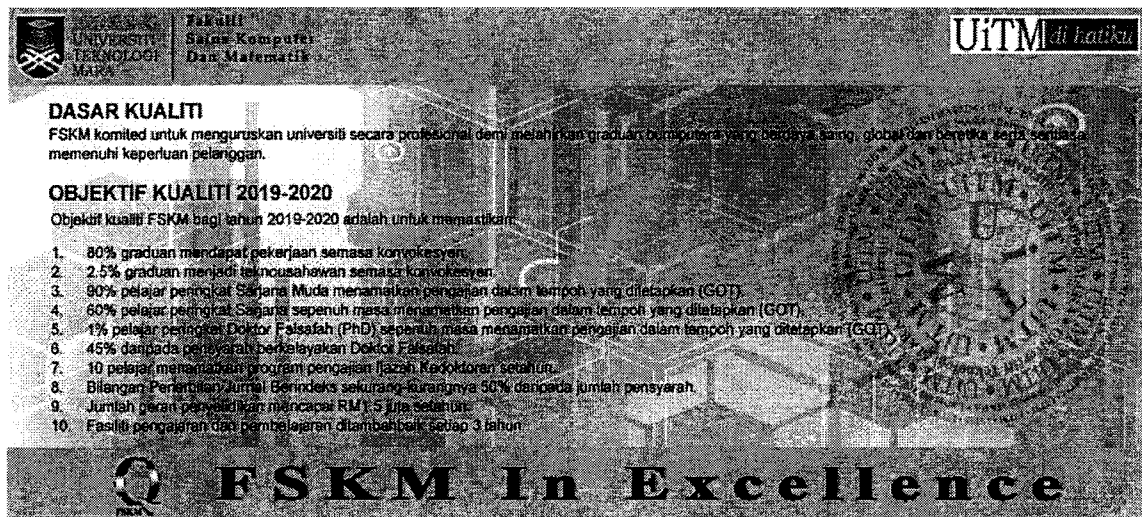


Figure 2.0 : FSKM Website

(8 marks)

b) Indicate the purpose of having entity integrity in a relational database.

(2 marks)

c) In database design, there are two classical approaches which consists of Top-down and Bottom-up. Differentiate these approaches.

(4 marks)

d) Present **THREE (3)** types of periodic maintenance in DBMS.

(6 marks)

QUESTION 3

Based on the relational schema used in Aprilbank Banking System, write SQL commands for the following questions.

BANK (bankCode, bankName, bankAddress)
 ACCOUNT (accNo, balance, accType, branchCode)
 LOAN (loanNo, amount, loanType, branchCode)
 CUSTOMER (custIC, custName, custAddress, branchCode)
 BRANCH (branchCode, branchAddress, bankCode)

a) Create ACCOUNT table.

(3 marks)

b) Customer phone number is important for this system. Add customer phone number attribute and named it as 'custPhone'.

(2 marks)

c) Find total number of customers who opened their account at Aprilbank.

(2 marks)

- d) Find the list of customers who has more than RM 30,000 in his/her account at Melaka branch. (3 marks)
- e) Display customers whose loan type is either Personal or Housing. Sort the results based on the account type ascending. (3 marks)
- f) Display names of customers who have an account at the bank, and whose names are neither 'Intan' nor 'Ahmad'. (3 marks)
- g) Find total amount of personal loan for each branch. (4 marks)

QUESTION 4

Based on order form shown in Figure 3.0, answer the following questions:

DORAEMON TOYS PARK & TRADING 2, Jalan Bunga Raya 7, Taman Tasik Jaya, 70400 Seremban, Negeri Sembilan Malaysia					
Customer No:	153	INVOICE			
Customer Name:	Puan Adriana Yusuf	No:	2088		
Customer Address:	Taman Rasah Jaya, Seremban	Date:	26 August 2019		
Item No	Item Name	Model	Quantity	Item Price	Total
CRB101	Electric Kids Car	Buggati	1	559.00	559.00
LEG101	City Fire Ladder Truck	LEGO	1	184.00	184.00
LEG102	City ATV Race Team	LEGO	2	79.99	159.98
Total Amount:					902.95

Figure 3.0: Customer invoice for Doraemon Toys Park & Trading

- a) Convert the data in Figure 3.0 into a relational schema format. (2 marks)
- b) State the primary key(s) based on (a). (2 marks)
- c) Normalize the table until third normal form (3NF). Show all the steps taken. (16 marks)

QUESTION 5

Auto Servis Sdn Bhd launched their first ever automotive company. Auto Servis Sdn Bhd plan to hire a database designer to build their own Car Dealership System (CDS). These requirements are required to model the database of CDS :

Each salesperson is managed by one manager and each manager needs to manage many salespersons. Manager has manager ID, first name and last name. A salesperson has their ID, first name and last name. A salesperson is assigned to more than one customer. Each customer has their customer ID, first name, last name, phone number, address, city, state and country. Each customer can buy more than one vehicle. Every vehicle has its own, unique Vehicle Identification Number (VIN) which also referred to as the body number. Each vehicle can be categorized as Car, Truck and SUV. Car has engine size, truck has tonnage and SUV has number of seats as their attribute respectively. An invoice is issued to customers. It includes a unique invoice number, a salesperson's ID, VIN as well as the date of invoice.

Based on the given information :

- a) Construct a complete Entity Relationship Diagram (ERD) with all required entities, attributes, relationships as well as the cardinalities using the Crow's Foot notation and include relevant attributes when necessary.

(16 marks)

- b) Briefly describe **TWO (2)** examples of reports that can be generated from ERD in (a).

(4 marks)

END OF QUESTION PAPER